Table 1: Overall contact fidelity and dynamic quality on CORE4D-G1 (52 sequences / 11 objects; ankle jerk omitted). ↑: higher is better, ↓: lower is better. **Bold** = best. Ours = SPIDER-CEM. Contact metrics are penetration-aware (a loose "any-contact" count that rewards interpenetration is intentionally omitted). Dynamic-quality metrics for Ours and OmniRetarget are computed on a MuJoCo physics rollout (native for Ours; position-servo replay for the kinematic OmniRetarget, E197 protocol) via the same code, hence directly comparable.

Metric	Ours	OmniRt
Phys. contact@3mm ↑	**0.588**	0.101
Phys. contact@10mm ↑	**0.804**	0.510
Phys. pen@3mm ↓	**0.176**	0.536
Phys. pen@10mm ↓	**0.017**	0.247
Max phys pen (mm) ↓	**12.3**	21.1
Geom. pen@2mm ↓	**0.052**	0.381
Obj speed max (m/s) ↓	**1.29**	2.83
Foot slip max (m) ↓	**0.87**	1.33